

EDUGENIE — COMPLETE PROJECT DOCUMENTATION

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1. PROJECT OVERVIEW

EduGenie is a lightweight AI-powered educational assistant built with FastAPI and Google Gemini. It offers five features: Q&A, explanations, summarization, quiz generation, and learning paths, all inside a single-page web interface.

2. PROBLEM STATEMENT

Students lack affordable and quick learning support. Tutoring costs too much, search engines return unreliable results, and existing AI tools require paid subscriptions. EduGenie solves this with a free, fast, distraction-free AI study companion.

3. TECHNICAL ARCHITECTURE

[Browser]

↓ Fetch API

[FastAPI backend]

↓ Python function calls

[Google Gemini (gemini-3.8-flash)]

↓ JSON response

[Output box in browser]

All 5 features share a single helper function ``_gen(prompt)`` for

minimal code duplication.

4. KEY FEATURES

Feature 1: Q&A — instant answers to any academic question

Feature 2: Explanation — simplified topic descriptions

Feature 3: Summarization — condense long passages

Feature 4: Quiz Generator — 3 MCQs with 4 options

Feature 5: Learning Paths — structured roadmaps with resources

5. TECH STACK

Python 3.10+, FastAPI, Jinja2, Google Gemini (gemini-3.8-flash),

HTML5, CSS3, Vanilla JavaScript, python-dotenv.

6. RESULTS

All 10 test cases passed. Average response time is 8–10 seconds.

The application runs entirely free with zero infrastructure cost.

7. FUTURE SCOPE

- Voice input for hands-free learning**
- Multi-language support (Hindi, Tamil, Spanish)**
- Progress tracking dashboard**
- Integration with Google Classroom and Moodle**
- Offline mode with cached answers**
- Mobile app (React Native)**
- Image and PDF input support for OCR-based learning**

8. CONCLUSION

EduGenie demonstrates how Generative AI can democratize education by providing every student with a free, fast, and reliable AI tutor. Its simple architecture and clean design make it a solid foundation for a scalable educational platform.